

Austin Song

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EDUCATION

University of Virginia

M.S. in Computer Science

Bachelor's in Computer Science (Cybersecurity Focal Path), GPA: 3.90

Charlottesville, VA

May 2028

May 2027

EXPERIENCE

Software Engineering Intern | Capital One | Node.js, Python, Jenkins

May 2026 – Aug 2026

- Built Node.js backend integrations for a company-wide network security intake platform, consolidating 6 Jira workflows for firewall, WAF, proxy, bot, and DDoS requests.
- Hardened request handling across intake workflows with server-side validation, preventing malformed inputs from reaching downstream services and closing 7 identified validation gaps.
- Analyzed 1,600+ historical Jira threads to identify recurring requester-engineer friction, then redesigned workflow requirements adopted into the intake teams' permanent forms.

Software Engineering Intern | Johns Hopkins APL | Java, Jython

May 2025 – Aug 2025

- Engineered a dynamic dependency-graph system across 1,000+ nodes, using out-degree centrality and configurable alert thresholds to surface mission-critical assets for external planners.
- Implemented runtime X.509 certificate-chain validation and dynamic trust-store management across 7+ services, improving secure service-to-service communication and detecting untrusted certificates.
- Developed a sandboxing layer for a Jython scripting environment by restricting reflection, dynamic imports, and privileged built-ins to prevent unauthorized system access.

LLM Researcher | DART Lab, University of Virginia | Python, OpenAI API

May 2026 – Present

- Built LLM agents to audit SIGMA, Elastic, and Splunk detection rules, using confidence scoring and cited threat intelligence to identify and rank inconsistencies.

PROJECTS

Job Scheduler | TypeScript, gRPC, PostgreSQL

Feb 2026

- Built a distributed job scheduler sustaining 2,000+ jobs/sec across concurrent workers using gRPC and PostgreSQL.
- Increased median throughput by 45% through batched result reporting, reducing RPC overhead under sustained load.
- Implemented renewable leases, retries, and token validation to recover failed jobs and reject stale worker results.

DiffSentry | FastAPI, GitHub Actions, Docker, Python, Cloudflare Analytics

Mar 2025

- Engineered a GitHub Actions-integrated static-analysis pipeline for continuous vulnerability scanning, achieving 90%+ detection accuracy with a 3% false-positive rate.
- Automated multi-stage analysis workflows covering the OWASP Top 10 and 40+ common C vulnerabilities to detect injection, XSS, access-control, and memory-safety flaws.

SELECTED RESEARCH & HONORS

- 2× CCDC National Finalist & Mid-Atlantic Champion (2025, 2026):** Won two national silver medals and two gold regional medals for UVA in the Collegiate Cyber Defense Competition.
- NSF REU (2026):** Selected for a \$10,000 research stipend supporting in-context reinforcement-learning research.
- University of Chicago NOISE Lab (2024):** Co-authored a [published paper](#) evaluating LLM behavior in mental-health contexts.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL

Backend/Systems: Ansible, Terraform, FastAPI, Node.js, REST APIs, gRPC, PostgreSQL, MySQL, Docker, Linux

Developer Tools: Git, GitHub Actions, Jenkins, CI/CD

ML/Data: PyTorch, TensorFlow